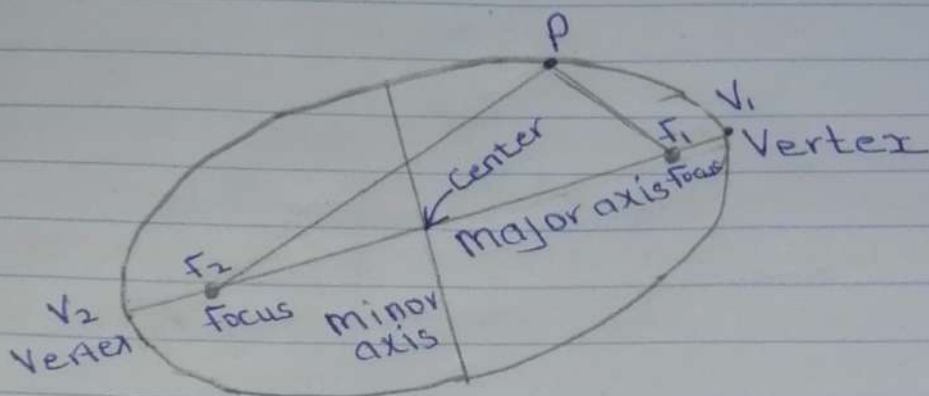


The ellipse:

(119)

An ellipse is the set of points in a plane the sum of whose distances from two fixed points in the plane is a constant.



The two fixed points are called the foci of the ellipse and the midpoint of the line segment joining the foci is called the center.

For any point P on the ellipse, the sum $PF_1 + PF_2$ is a constant, where F_1 and F_2 are the foci.

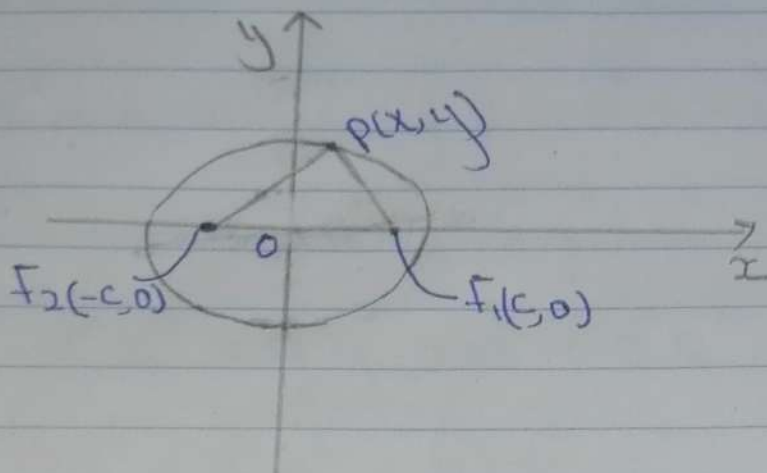
The points V_1 and V_2 where the ellipse crosses the line through the foci are called vertices of the ellipse.

The chord of the ellipse joining the vertices is called the major axis, and the chord through the center perpendicular to the major axis is called the minor axis.

The simplest equations of ellipse result when the center is at the origin and the major axis lies on one of the coordinate axes.

(120)

Let's consider the equation of an ellipse when the major axis lies on the x-axis.



A point $P(x,y)$ lies on the ellipse if and only if the sum of the distances $\overline{PF_1}$ and $\overline{PF_2}$ is a constant.

Suppose the constant $= 2a$

Notice that we must have $a > c$.

$$\Rightarrow \overline{PF_1} + \overline{PF_2} = 2a$$

$$\sqrt{(x-c)^2 + y^2} + \sqrt{(x+c)^2 + y^2} = 2a$$

$$\sqrt{(x-c)^2 + y^2} = 2a - \sqrt{(x+c)^2 + y^2}$$

Squaring on both sides

$$(x-c)^2 + y^2 = (2a - \sqrt{(x+c)^2 + y^2})^2$$

$$x^2 - 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + x^2 + 2cx + c^2 + y^2$$

$$4a\sqrt{(x+c)^2 + y^2} = 4a^2 + 4cx$$

$$a\sqrt{(x+c)^2 + y^2} = a^2 + cx$$

(121)

Squaring on both sides.

$$a^2((x+c)^2 + y^2) = a^4 + 2a^2cx + c^2x^2$$

$$a^2(x^2 + 2cx + c^2 + y^2) = a^4 + 2a^2cx + c^2x^2$$

$$a^2x^2 + 2a^2cx + a^2c^2 + a^2y^2 = a^4 + 2a^2cx + c^2x^2$$

$$a^2x^2 - c^2x^2 + a^2y^2 = a^4 - a^2c^2$$

$$(a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$$

Dividing all through with $a^2(a^2 - c^2)$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1$$

Since $a > c$, we can simplify further by writing

$$b^2 = a^2 - c^2 : b > 0$$

obtaining

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \dots \dots \textcircled{*}$$

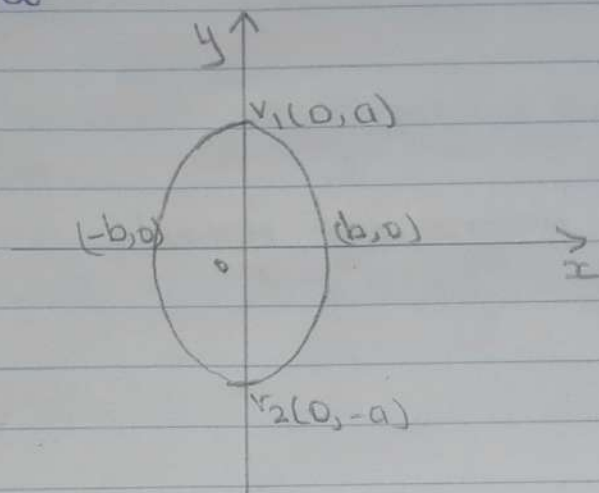
Since $c > 0$ and $b^2 = a^2 - c^2$ it follows that $a^2 > b^2$ and hence $a > b$.

$\textcircled{*}$ refer to the standard form of the equation with center at the origin and horizontal major axis

(122)

When the major axis is on the y-axis, a similar analysis gives the standard form

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1.$$



NB: Length of major axis = $2a$
Length of minor axis = $2b$

Remark: The position is determined by the sizes of the denominators when the equation is in standard form. Whichever denominator is larger is a^2

$\Rightarrow$ for the equation

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\Rightarrow a=4 \quad \text{and} \quad b=3.$$

With the major axis along the x-axis

Standard forms of Equations of Ellipses. (123)

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, \text{ major axis horizontal}$$

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1, \text{ major axis vertical.}$$

$\Rightarrow ax^2 + by^2 + cx + dy + e = 0$, Where a and b are like in sign but are unequal.

Examples

1. Discuss and sketch the graph of the equation
Solution $4x^2 + 18y^2 = 36$.

$$4x^2 + 18y^2 = 36$$

Dividing all through with 36, we obtain

$$\frac{x^2}{9} + \frac{y^2}{2} = 1$$

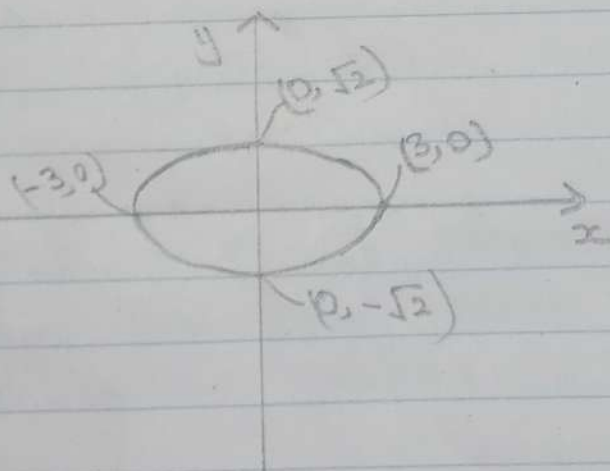
$$\Rightarrow a^2 = 9 \Rightarrow a = 3$$

$$b^2 = 2 \Rightarrow b = \sqrt{2}$$

The endpoints of the major axis are $(\pm 3, 0)$
and the endpoints of the minor axis are $(0, \pm \sqrt{2})$

$$\text{Since } c^2 = a^2 - b^2 = 9 - 2 = 7 \text{ or } c = \sqrt{7}$$

$\Rightarrow$ the foci are $(\pm \sqrt{7}, 0)$



(124)

2. Find an equation of the ellipse with Vertices $(\pm 4, 0)$ and Foci $(\pm 2, 0)$

Solution

From this information $a=4$ and $c=2$.

But $c^2 = a^2 - b^2$

$$4 = 16 - b^2 \Rightarrow b^2 = 12.$$

The major axis is along x -axis

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{12} = 1$$

$$\Rightarrow 3x^2 + 4y^2 = 48.$$

3. ~~4x^2~~ Discuss the graph of the equation

$$4x^2 + 9y^2 + 16x - 18y - 11 = 0.$$

Solution

$$4x^2 + 9y^2 + 16x - 18y - 11 = 0$$

$$4x^2 + 16x + 9y^2 - 18y = 11.$$

$$4(x^2 + 4x) + 9(y^2 - 2y) = 11$$

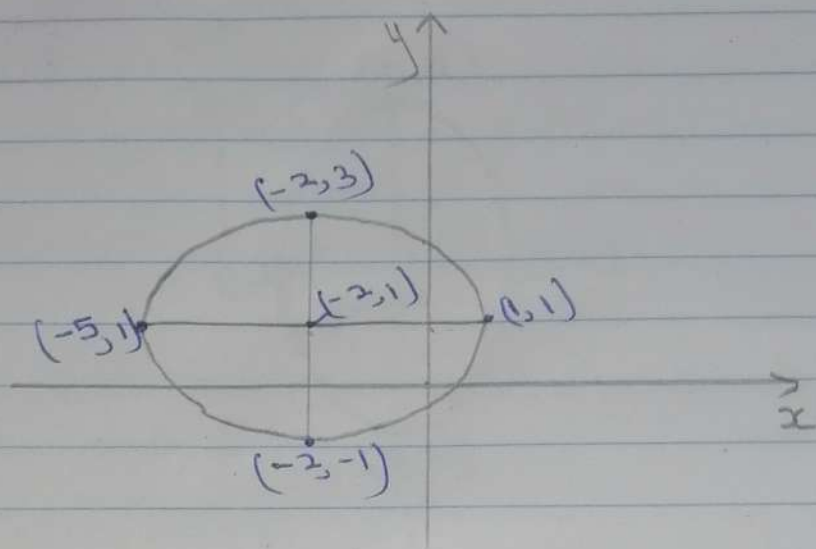
$$4(x^2 + 4x + 4) + 9(y^2 - 2y + 1) = 11 + 16 + 9$$

$$4(x+2)^2 + 9(y-1)^2 = 36.$$

$$\frac{(x+2)^2}{9} + \frac{(y-1)^2}{4} = 1.$$

$\Rightarrow$ An equation of an ellipse with $a=3$, $b=2$ and major axis being horizontal with center $(-2, 1)$

(125)



4. Discuss and sketch the graph of the equation $9x^2 + 4y^2 - 54x - 16y + 61 = 0$.

Solution

$$9x^2 + 4y^2 - 54x - 16y + 61 = 0$$

$$9x^2 - 54x + 4y^2 - 16y = -61$$

$$9(x^2 - 6x) + 4(y^2 - 4y) = -61$$

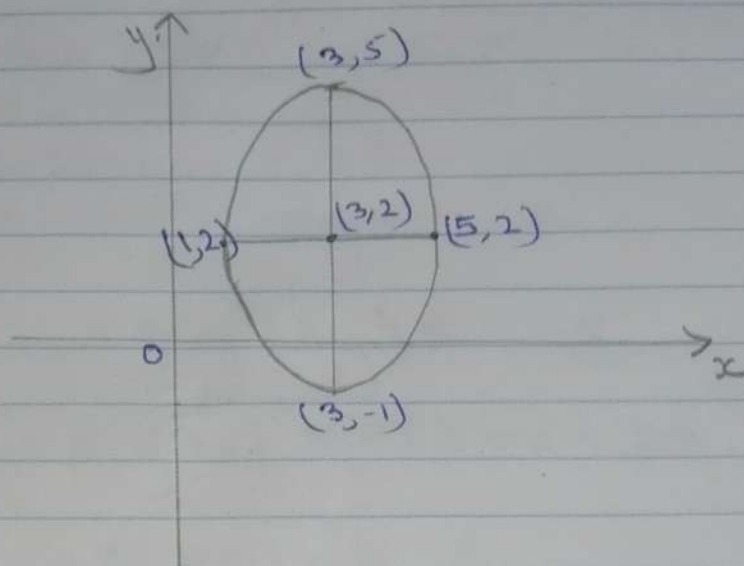
$$9(x^2 - 6x + 9) + 4(y^2 - 4y + 4) = -61 + 81 + 16$$

$$9(x-3)^2 + 4(y-2)^2 = 36$$

$$\frac{(x-3)^2}{4} + \frac{(y-2)^2}{9} = 1$$

$\Rightarrow$ Equation of an ellipse centered at $(3, 2)$, $a=3$ and $b=2$.

The major axis is along the y -axis.



5. Discuss the graph of the equation
 $4x^2 + 5y^2 - 8x + 20y + 24 = 0$.

Solution

$$4x^2 + 5y^2 - 8x + 20y + 24 = 0$$

$$4x^2 - 8x + 5y^2 - 20y = -24$$

$$4(x^2 - 2x) + 5(y^2 - 4y) = -24$$

$$4(x-1)^2 + 5(y+2)^2 = -24 + 4 + 20$$

$$4(x-1)^2 + 5(y+2)^2 = 0$$

Since the right-hand side is 0, this cannot be put in standard form.

The equation is satisfied only by the single point $(1, -2)$.

We call this a degenerate ellipse.

6. Discuss the graph the equation
 $4x^2 + 5y^2 - 8x + 20y + 25 = 0.$

Solution

$$4x^2 - 8x + 5y^2 + 20y = -25.$$

$$4(x^2 - 2x) + 5(y^2 + 4y) = -25.$$

$$4(x-1)^2 + 5(y+2)^2 = -25 + 4 + 20$$

$$4(x-1)^2 + 5(y+2)^2 = -1.$$

Since the left-hand side is always nonnegative, the equation is not satisfied by any point, and hence there is no graph.

Exercise:

1. Find the standard form of the equation of an ellipse with the given properties

(a) x-intercepts ± 4 ; y-intercepts ± 2 .

(b) Foci at $(-2, 0)$ and $(-2, 8)$; one vertex at $(-2, -1)$

(c) End points of minor axis at $(4, -3)$ and $(4, -7)$; one focus at $(8, -5)$

2. Write the equation in standard form, and sketch the graph.

(a) $25x^2 + 4y^2 = 100$

(b) $5x^2 = 3(1 - 5y^2)$

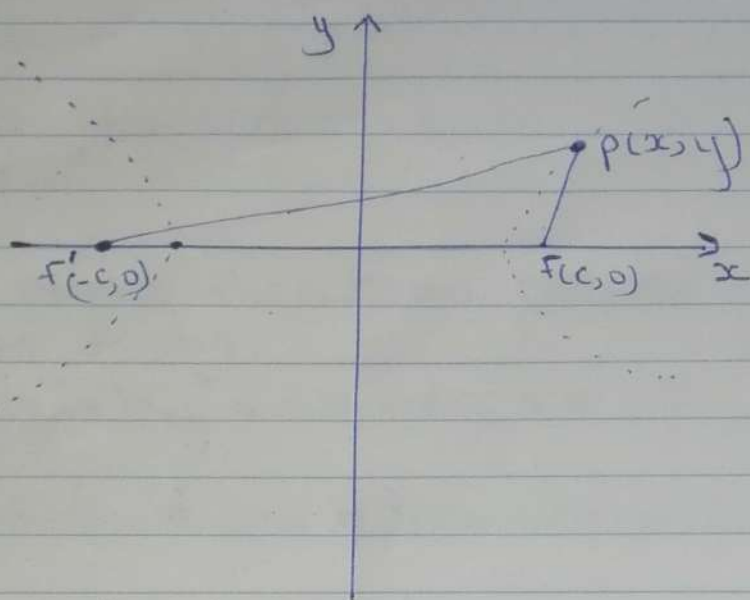
(c) $x^2 + 4y^2 - 2x + 16y + 13 = 0$

(d) $2x^2 + 4y^2 - 5x + 6y - 4 = 0$

The hyperbola.

(128)

A hyperbola is the set of all points in a plane, the difference of whose distances from two fixed points in the plane (the foci) is a positive constant.



$$\Rightarrow d(P, F') - d(P, F) = 2a$$

For hyperbolas (unlike ellipses) we need $a < c$ in order to obtain points on the hyperbola.

1. The graph of the equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

is a hyperbola with vertices $(\pm a, 0)$.

The foci are $(\pm c, 0)$, where $c^2 = a^2 + b^2$.

2. The graph of the equation $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ is a hyperbola with

vertices $(0, \pm a)$. The foci are $(0, \pm c)$, where $c^2 = a^2 + b^2$.

In both cases, center is at the origin.

(129)

Standard forms of Equations of Hyperbola.

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1, \text{ transverse axis horizontal}$$

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1, \text{ transverse axis vertical}$$

Where the center is at (h, k) and $b^2 = c^2 - a^2$.

The above equation can be written in the form

$$ax^2 + by^2 + cx + dy + e = 0$$

in which a and b are opposite in sign.

Exercise

1. Find the equation in standard form of the hyperbola that satisfies the following conditions. Sketch the graph.

(a) Center at $(-1, 3)$; one focus at $(4, 3)$; one vertex at $(-4, 3)$

(b) Vertices at $(2, -2)$ and $(2, 4)$, eccentricity $= \sqrt{5}$

2. Discuss and sketch the graph of

(a) $4x^2 - 9y^2 - 16x - 18y - 29 = 0$

(b) $x^2 - 4y^2 + 4x + 2y - 28 = 0$.

(c) $16x^2 - 25y^2 - 64x + 200y - 336 = 0$.

(d) $9x^2 - 16y^2 + 36x + 96y - 108 = 0$.

(130)

3. Find the equations of the tangent line and normal line to the hyperbola $3x^2 - y^2 + 6x + 4y + 8 = 0$ at the point $(2, -4)$.

4. Find equations of the tangent lines to the hyperbola $8x^2 - 3y^2 = 48$ that are parallel to the line $2x - y = 10$.